Assignment 6

(Due date: 07-04-2026)

Hand calculations

1. (20 points) Consider solving the following first-order initial value problem:

$$y'(x) = \frac{dy}{dx} = f(x, y) = x + y, \quad y(0) = 1, \quad 0 \le x \le 1,$$

using a fourth-order accurate Taylor-series method given as follows:

$$y(x_i + h) = y(x_i) + h\, T_4(x_i, y_i, h)$$

where

$$T_4(x_i, y_i, h) = y'(x_i) + y''(x_i)\frac{h}{2!} + y'''(x_i)\frac{h^2}{3!} + y^{iv}(x_i)\frac{h^3}{4!}.$$

Starting with $x = 0$ using a step size of $h = 0.1$, calculate the solution at $x = 1$. Tabulate your results showing the computed solution, exact solution and error at each step. The exact analytical solution is given by $y(x) = 2e^x - x - 1$.

Programming

1. (60 points) Consider the following initial value problem

$$\frac{dy}{dt} = -\lambda\,(y - e^{-t}) - e^{-t}, \quad \lambda > 0$$

with the intial condition $y(t = 0) = y_0$. Solve this initial value problem for $\lambda = 10$ and $y_0 = 10$ using explicit Euler, implicit Euler, trapezoidal, second-order Runge-Kutta, fourth-order Runge-Kutta and Runge-Kutta-Fehlberg method. Use time step sizes $h = 0.1, 0.05, 0.025, 0.0125, 0.00625$ and integrate from $t = 0$ to 0.8. The exact analytical solution to this equation is

$$y = e^{-t} + (y_0 - 1)e^{-\lambda t}.$$

(a) For each method, plot the solutions obtained along with the exact solution from $t = 0$ to $t = 0.8$.
(b) Evaluate the error between the numerical and the exact solution at $t = 0.8$ as follows. If the order of the method is p, then the ratio of the error corresponding to a step size of h and $h/2$ should be 2^p. Tabulate this ratio for successive pairs of h and deduce the asymptotic value of p for each method.
Use double precision arithmetic throghout. The formulation of the methods is given as follows,

- Explicit Euler Method

$$y_{i+1} = y_i + hf(x_i, y_i)$$

- Implicit Euler Method

$$y_{i+1} = y_i + hf(x_{i+1}, y_{i+1})$$

- Trapezoidal Method

$$y_{i+1} = y_i + \frac{h}{2}\left[f(x_i, y_i) + f(x_{i+1}, y_{i+1})\right]$$

- Second-Order Runge-Kutta Method

$$y_{i+1} = y_i + hk_2$$

where

$$k_1 = f(x_i, y_i)$$
$$k_2 = f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right)$$

- Fourth-Order Runge-Kutta Method

$$y_{i+1} = y_i + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

where

$$\begin{aligned}
k_1 &= f(x_i, y_i) \\
k_2 &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right) \\
k_3 &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right) \\
k_4 &= f(x_i + h, y_i + hk_3)
\end{aligned}$$

- Runge-Kutta-Fehlberg Method

$$y_{i+1} = y_i + \left(\frac{16}{135}k_1 + \frac{6656}{12825}k_3 + \frac{28561}{56430}k_4 - \frac{9}{50}k_5 + \frac{2}{55}k_6\right)$$

where

$$\begin{aligned}
k_1 &= hf(x_i, y_i) \\
k_2 &= hf\left(x_i + \frac{h}{4}, y_i + \frac{k_1}{4}\right) \\
k_3 &= hf\left(x_i + \frac{3h}{8}, y_i + \frac{3k_1}{32} + \frac{9k_2}{32}\right) \\
k_4 &= hf\left(x_i + \frac{12h}{13}, y_i + \frac{1932k_1}{2197} - \frac{7200k_2}{2197} + \frac{7296k_3}{2197}\right) \\
k_5 &= hf\left(x_i + h, y_i + \frac{439k_1}{216} - 8k_2 + \frac{3680k_3}{513} - \frac{845k_4}{4104}\right) \\
k_6 &= hf\left(x_i + \frac{h}{2}, y_i - \frac{8k_1}{27} + 2k_2 - \frac{3544k_3}{2565} + \frac{1859k_4}{4104} - \frac{11k_5}{40}\right).
\end{aligned}$$